The Gravitational Dyon Pair: Quarter Quantisation on the Poincaré Cone

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Abstract

A mass m and a NUT charge l form gravity's dyon pair: the monopole of the gravitoelectric field and the monopole of the gravitomagnetic field. That the pair obeys a Dirac-type quantisation condition has been known in essence since Misner and Dowker, and is usually left as a remark. This note pins the condition down. From the SU(2) Killing triad of Taub–NUT we show that a test particle's conserved angular momentum satisfies $\mathbf{J}\cdot\hat{\mathbf{n}} = -2\varepsilon l$ exactly, so the pair's stored angular momentum is $2\varepsilon l$, twice the electromagnetic value at the same formalism; the Misner time period $8\pi l$ gives the same lattice by an independent route. The pair product is therefore quantised in quarters, $\mu c l/\hbar \in \frac{1}{4}\mathbb{Z}$, i.e. $\alpha_G(\mu)\,\alpha_G(\tilde{\mu}) = (k/4)^2$ with minimum $1/16$, against the electromagnetic $\alpha\,\alpha_m = (1/2)^2 = 1/4$. We then exhibit structures of the pair that appear not to have been stated: its seats in the parabolic separation of the MIC–Kepler problem are $k/2 \pm \varepsilon l$; the monopole enters the spectrum only through the offset $\Lambda + \frac{1}{2} = \sqrt{(j+\frac{1}{2})^2 - s^2}$ of the ladder's diameter–secant split; the electromagnetic half-lattice is the even sublattice of the gravitational quarter-lattice; and a boost decomposes the state's Casimir between its orbital and pair legs, saturating at a finite rapidity $\gamma_{\max} = 2\sqrt{j(j+1)}/n$. All identities were verified by exact symbolic computation.

1 Two dyon pairs

Electromagnetism has one observed monopole, the electric charge, and one exact zero, $\nabla\cdot\mathbf{B} = 0$. Gravity repeats the pattern one column over: mass is the observed monopole of the gravitoelectric field, and the monopole of the gravitomagnetic field—the NUT charge, or dual mass—is unobserved. In each theory the two charges form a dyon pair, and in each theory the pair is constrained by single-valuedness.

For the electromagnetic pair the constraint is Dirac's [1]. Its sharpest derivation is Poincaré's: the field of a charge–monopole pair stores angular momentum eg/c, directed along the line joining them, independently of their separation. The total angular momentum of a test charge therefore has a fixed radial component,

$$\mathbf{J}\cdot\hat{\mathbf{n}} = -eg, \qquad J^2 = L^2 + (eg)^2, \tag{1.1}$$

with $L \perp \hat{\mathbf{n}}$ the orbital part; the position direction $\hat{\mathbf{n}}$ is confined to a cone about $\mathbf{J}$ (the Poincaré cone), and SU(2) consistency forces $eg \in \frac{\hbar}{2}\mathbb{Z}$. In dimensionless couplings, with $\alpha = e^2/\hbar c$ and $\alpha_m = g^2/\hbar c$,

$$\alpha\,\alpha_m = \left(\frac{eg}{\hbar c}\right)^2 = \left(\frac{k}{2}\right)^2, \qquad k \in \mathbb{Z}, \tag{1.2}$$

with minimum $1/4$: the pair product is the square of the pair's angular-momentum quantum, $\hbar/2$.

The gravitational analogue—Taub–NUT [2, 3] read as a gravitational dyon—goes back to Demiański–Newman and to Dowker [6], and the existence of a quantisation condition relating mass and NUT charge, obtained from the periodicity of the time coordinate, is recorded e.g. in the double-copy literature [7] following Misner [4]. What seems not to be in print is the condition's coefficient in pair-product form, nor the structures of Sections 5–8. We supply both routes to the coefficient here because their agreement is the content: the factor of two that decides between $1/4$ and $1/16$ is the fibration's own 2, read once from particle dynamics and once from topology.

2 The pair term from the Killing triad

Write Taub–NUT in the standard parametrisation, in geometric units, with $\Sigma = r^2 + l^2$ and $f(r) = (r^2 - 2mr - l^2)/\Sigma$:

$$ds^2 = -f\,\chi^2 + \frac{dr^2}{f} + \Sigma\,(d\theta^2 + \sin^2\theta\,d\varphi^2), \qquad \chi = dt + 2l\cos\theta\,d\varphi. \tag{2.1}$$

The isometry group is $\mathrm{SU}(2) \times \mathrm{U}(1)$. The SU(2) factor is not the naive rotation group of the (θ, φ) sphere: each would-be rotation must be accompanied by a t-shift to remain a symmetry. Solving the Killing equation with $f(r)$ left arbitrary, the shift is forced (the opposite sign fails), giving the triad

$$\xi_x = 2l\,\frac{\cos\varphi}{\sin\theta}\,\partial_t - \sin\varphi\,\partial_\theta - \cos\varphi\cot\theta\,\partial_\varphi, \tag{2.2}$$

$$\xi_y = 2l\,\frac{\sin\varphi}{\sin\theta}\,\partial_t + \cos\varphi\,\partial_\theta - \sin\varphi\cot\theta\,\partial_\varphi, \tag{2.3}$$

$$\xi_z = \partial_\varphi, \tag{2.4}$$

which closes as an $\mathfrak{su}(2)$ algebra, $[\xi_i, \xi_j] = -\epsilon_{ijk}\xi_k$. The closure is a real constraint: the alternative third generator $\partial_\varphi + 2l\,\partial_t$ is also a Killing vector, being a linear combination of symmetries, but it does *not* close the algebra with ξ_x, ξ_y.

For a unit-mass geodesic let $\varepsilon = -p_t$ (specific energy) and $k = p_\varphi$. The conserved charges $Q_i = \xi_i \cdot p$ assemble into a total angular momentum $\mathbf{J}$ satisfying two exact identities:

$$\mathbf{J}\cdot\hat{\mathbf{n}} = -2\varepsilon l, \qquad J^2 = \Sigma^2\dot\theta^{\,2} + \frac{\big(k + 2\varepsilon l\cos\theta\big)^2}{\sin^2\theta} + \big(2\varepsilon l\big)^2. \tag{2.5}$$

This is Poincaré's structure (1.1) with the replacement

$$eg \;\longrightarrow\; 2\varepsilon l. \tag{2.6}$$

Three remarks. First, the "electric" member of the gravitational pair is the particle's *energy*: the gravitomagnetic charge couples to ε, as the Kaluza–Klein reading of the t-fibration requires. At rest at infinity $\varepsilon = 1$. Second, $\mathbf{J}\cdot\hat{\mathbf{n}}$ fixed confines $\hat{\mathbf{n}}$ to a cone about $\mathbf{J}$ with $\cos\alpha = -2\varepsilon l/|\mathbf{J}|$: the Poincaré cone, in the gravitational theory, with the pair term as one leg of the cone relation $J^2 = (\text{orbital})^2 + (\text{pair})^2$. Third, the coefficient is 2, not 1: it is the 2 in $\chi = dt + 2l\cos\theta\,d\varphi$, and it is physical, since the same l appears in f and Σ and admits no reparametrisation.

3 The Misner route

The same coefficient controls the string. Near the poles, $\chi \approx dt \pm 2l\,d\varphi$; the regular time coordinates $t_N = t + 2l\varphi$ and $t_S = t - 2l\varphi$ disagree by $4l\varphi$ on the overlap, hence by $8\pi l$ after one wind. Removing

the Misner string therefore requires t periodic with period $T_t = 8\pi l/c$ [4], and single-valuedness of $e^{-iEt/\hbar}$ quantises energy in steps

$$E_k = \frac{2\pi\hbar}{T_t}\,k = \frac{\hbar c}{4l}\,k, \qquad k \in \mathbb{Z}. \tag{3.1}$$

Setting $E = \mu c^2$ places a test mass on the ladder. Both routes give the pair lattice of the next section, and both read the answer off the same 2: Route A as the strength of the fibration's monopole potential in the particle's angular momentum, Route B as the Misner period $8\pi l = 4\pi \cdot 2l$.

4 The pair product: quarters

Since $\mathbf{J}\cdot\hat{\mathbf{n}}$ is a component of an SU(2) generator and the remaining part of $\mathbf{J}$ is orthogonal to $\hat{\mathbf{n}}$, equation (2.5) with $\varepsilon = 1$ requires $2\mu_{\rm geo} l_{\rm geo}/\ell_P^2 \in \frac{1}{2}\mathbb{Z}$ for a test mass μ, i.e.

$$\frac{\mu_{\rm geo}\, l_{\rm geo}}{\ell_P^2} \in \frac{1}{4}\mathbb{Z} \qquad \Longleftrightarrow \qquad \frac{\mu\, c\, l}{\hbar} \in \frac{1}{4}\mathbb{Z}, \tag{4.1}$$

the second form showing that G cancels identically: the condition is dimensionless, G-free, and a property of the pair alone. With the dual mass $\tilde{\mu} = c^2 l/G$ and the dimensionless gravitational couplings $\alpha_G(\mu) = G\mu^2/\hbar c$, $\alpha_G(\tilde{\mu}) = G\tilde{\mu}^2/\hbar c$,

$$\boxed{\alpha_G(\mu)\,\alpha_G(\tilde{\mu}) = \left(\frac{k}{4}\right)^2, \qquad k \in \mathbb{Z}, \qquad \text{minimum } \frac{1}{16}.} \tag{4.2}$$

Gravity's pair quantum is $\hbar/4$, against electromagnetism's $\hbar/2$.

The exponent structure is worth displaying. The electromagnetic product is $(1/2)^2$: one factor $1/2$ from single-valuedness on the half-ladder, squared by the pairing. The gravitational product is $1/16 = (1/2)^{2\cdot 2}$: the same half, halved once more by the spin-2 fibration coefficient, then squared by the pairing. The quarter is also the cone's native quarter: on the angular-momentum ladder the Casimir obeys $T^2 = j(j+1) + \frac{1}{4}$ with $T = j + \frac{1}{2}$, i.e. the offset between the arithmetic radius $j + \frac{1}{2}$ and the geometric Casimir $\sqrt{j(j+1)}$ is $1/2$, and its square—the deficit in T^2—is $1/4$. The pair quantum $\hbar/4$ and the Casimir deficit $1/4$ are the same quarter seen in two places: one in a product of charges, one in a spectrum. The mechanism has been isolated before [13] as *dyadic quadratic normalisation*: a canonical half-coordinate becomes a quarter under a quadratic operation, with four exact realisations (the factor-cell midpoint square, the shifted compact and noncompact Casimirs, the primitive null-edge midpoint norm, and the cyclotomic relative-norm normalisation). The pair product (4.2) is that mechanism at dyadic depth two—half to quarter for the pair quantum, quarter to sixteenth for its square—and its first realisation in a product of charges. The same square root governs the power laws on the parabolas themselves: a parabolic segment has area $\propto h^{3/2}$ with $3/2 = 1 + \frac{1}{2}$, and the bouncer spectrum warps as $E \propto (n - \frac{1}{4})^{2/3}$ with $2/3 = 1/(1 + \frac{1}{2})$ [14].

5 Parabolic seats

The Kepler problem separates in parabolic coordinates $\xi = r(1 + \cos\theta)$, $\eta = r(1 - \cos\theta)$, and its parabolic quantum numbers are the seats from which the hydrogenic fine-structure bookkeeping is read off. The monopole deformation of Kepler—the MIC problem [8, 9]—is the flat-space twin of Taub–NUT geodesic motion. Its Hamilton–Jacobi equation separates in the same coordinates.

Exactly (including the Zwanziger term $+s^2/r^2$, without which a remainder $-s^2/(\xi+\eta)$ obstructs separation), with s the Poincaré strength and κ the Coulomb coupling:

$$\Big[\xi U'^2 + \frac{((k-s)/2)^2}{\xi} - \frac{E\xi}{2}\Big] + \Big[\eta V'^2 + \frac{((k+s)/2)^2}{\eta} - \frac{E\eta}{2}\Big] = \kappa. \tag{5.1}$$

Each parabola carries one charge, $(k \mp s)/2$: centre $k/2$, offset $s/2$. Hydrogen is $s = 0$, both seats equal to $(k/2)^2$.

Taub–NUT's Hamilton–Jacobi equation has MIC's angular and pair sectors *identically*, at $s = 2\varepsilon l$: subtracting the two equations,

$$\text{TN} - \text{MIC} = -(2mr + l^2)\, p_r^2 - (2\varepsilon l)^2, \tag{5.2}$$

which is purely radial—a p_r^2 weight and a constant, the sector absorbed by the standard time reparametrisation identifying the two flows [10]. The parabolic content is undeformed. Hence Taub–NUT's parabolic seats are

$$\frac{k}{2} - \varepsilon l, \qquad \frac{k}{2} + \varepsilon l : \tag{5.3}$$

the diameter is k, the arithmetic centre is $k/2$, and the offset is the pair product εl itself. The twist does not build a new lattice; it splits each hydrogen seat symmetrically about its centre by the pair term of Section 2. The state lattice of the separation thus runs in halves, while the pair lattice of Section 4 runs in quarters—one additional power of two of resolution, supplied by the fibration.

6 The monopole is the offset

The identity (2.5) says $J^2 = L^2 + s^2$ with s the pair strength ($s = eg$ electromagnetically, $s = 2\varepsilon l$ gravitationally): the orbital part is what remains of the Casimir after the pair takes its leg. Quantising, the mechanical angular momentum of the MIC system has eigenvalues $\Lambda(\Lambda+1)$ with [8]

$$\Lambda(\Lambda+1) = j(j+1) - s^2, \qquad \text{i.e.} \qquad \Lambda + \frac{1}{2} = \sqrt{\Big(j+\frac{1}{2}\Big)^2 - s^2}\,, \tag{6.1}$$

and the bound spectrum is hydrogenic in the shifted rung, $E = -\kappa^2/(2N^2)$ with $N = n_r + \Lambda + 1$, recovering hydrogen at $s = 0$.

Equation (6.1) is a diameter–secant split [14]: the arithmetic mean is $j + \frac{1}{2}$, the offset is s, and the rung the spectrum actually climbs is the geometric mean $\sqrt{\text{AM}^2 - \text{offset}^2}$. A monopole therefore does not deform the ladder; it offsets it, entering through the offset slot alone. The parabolic seats (5.3) show the same entry in the other coordinate system: centre $k/2$ untouched, offset εl.

Consequently both monopoles obey the machine's power laws, not by generating them but by inheriting them: the laws belong to the parabola and its square root—segment area $\propto h^{3/2}$ with $3/2 = 1 + \frac{1}{2}$, spectral warp $(n - \frac{1}{4})^{2/3}$ with $2/3 = 1/(1 + \frac{1}{2})$, and the dyadic half-to-quarter normalisation of Section 4—and the offset is the only door a monopole has into that machine. In the diameter–secant picture the offset $(n-k)/2$ was always the free parameter, the position chosen by the secant. The pair gives that slot a physical occupant: the product of the pair's two charges. The question "what fixes the offset" and the question "what carries monopole charge, and how much" are the same question.

7 Intertwining of the two lattices

Equations (1.2) and (4.2) place the two theories' pair lattices on one grid:

$$\frac{eg}{\hbar c} \in \frac{1}{2}\mathbb{Z} \subset \frac{1}{4}\mathbb{Z} \ni \frac{\mu c l}{\hbar}. \tag{7.1}$$

The electromagnetic half-lattice is the *even sublattice* of the gravitational quarter-lattice: the minimal electromagnetic pair quantum $\hbar/2$ is the $k = 2$ rung of the gravitational lattice, $2 \times \hbar/4$. Electromagnetism occupies every second rung; the odd rungs belong to the twist sector alone. Two lattices, one grid—and the grid is fixed by single-valuedness in both theories, not fitted in either.

8 Boost decomposition

A boost along the symmetry axis acts on the pair through ε alone: k, l and $|\mathbf{J}|$ are unchanged, while a boost from rest sends $\varepsilon \to \cosh\psi$. The parabolic seats (5.3) split hyperbolically about their fixed centre,

$$\text{seats} = \frac{k}{2} \pm l\cosh\psi, \tag{8.1}$$

and the Casimir budget $J^2 = (\text{orbital})^2 + (2\varepsilon l)^2$ is transferred, at fixed J^2, from the orbital leg to the pair leg: the boost decomposes the state into the pair. Writing the pair fraction with both quantisations imposed, $\mu c l/\hbar = n/4$ and $|\mathbf{J}|^2/\hbar^2 = j(j+1)$, the transfer saturates when the orbital leg vanishes, $2\varepsilon l = |\mathbf{J}|/\mu$, at the finite rapidity

$$\gamma_{\max} = \frac{2\sqrt{j(j+1)}}{n}, \qquad \psi_{\max} = \operatorname{arcosh}\gamma_{\max}. \tag{8.2}$$

At the saturation the Poincaré cone has closed, $\hat{\mathbf{n}}$ locked antiparallel to $\mathbf{J}$. The minimal state $(j, n) = (\frac{1}{2}, 1)$ has $\gamma_{\max} = \sqrt{3}$, $\psi_{\max} = \operatorname{arcosh}\sqrt{3} \approx 1.146$, and rest-frame split pair 1/3 / orbital 2/3 (arithmetic of the minimal rung, recorded as such). Finiteness cuts the other way as well: a shell j admits at rest only the pair charges $n \le 2\sqrt{j(j+1)}$—$n = 1$ for $j = \frac{1}{2}$; $n = 1, 2$ for $j = 1$; $n = 1, 2, 3$ for $j = \frac{3}{2}$.

On the source side, the $\psi \to \infty$ image of a boost is the Aichelburg–Sexl shockwave [11]: a fully boosted field collapses onto the null cone as an impulsive wave. The pair-side saturation (8.2) is the finite, quantised counterpart for a test state: the cone closes before the boost diverges.

9 The action is the integration

None of the preceding structure floats free of an action: the action is the Hamilton–Jacobi integral itself. The characteristic function $W = U(\xi) + V(\eta)$ of Section 5 *is* Hamilton's action for the pair system, separated into its two parabolic pieces, and each quantisation encountered above is a statement about its periods. Route A fixes the angular period: $\mathbf{J}$ is the action variable of the SU(2) motion, and the pair term is the part of that period carried by the twist. Route B fixes the time period: single-valuedness of $e^{-iEt/\hbar}$ on the Misner cycle. The parabolic power laws are periods again—the Archimedean segment area $\propto h^{3/2}$ is the action of the linear-potential (bouncer) problem, whose Bohr–Sommerfeld periods, with their Maslov quarter, give the warp $E \propto (n - \frac{1}{4})^{2/3}$. The lattice of Section 4 is, in one phrase, the period lattice of the pair's action in units of $\hbar$: halves where the action is electromagnetic, quarters where it is gravitational.

This places the structure one rung below a field theory, not outside the theory business altogether. A worldline effective theory of the pair is built on precisely this action, and for such a theory the results above are exact period data—matching conditions it must reproduce: the quarter lattice (4.2), the seats (5.3) and the offset law (6.1), and the boost saturation (8.2). What is absent is the field layer: a Lagrangian density for the gravitoelectric and gravitomagnetic fields jointly, with an expansion organised on top of the worldline action. Section 10 returns to what that absence does and does not affect.

10 Scope

Nothing here detects the pair. NUT charge is unobserved; Taub–NUT without the periodic-time identification carries the Misner string, and with it, closed timelike curves. Equations (4.2), (5.3) and (8.2) are statements about the solution space of general relativity and the representation theory of its isometries, not about nature's inventory. Nor does the pair product derive G: l is a free charge, and the lattice constrains products, not values—exactly Dirac's situation, where the condition quantises eg given a monopole and leaves α free.

What the structure does supply is the *form* a gravitational selection principle would take if one exists in this family: dimensionless, G-free once written as (4.1), and pair-dependent at once, with its symmetry—the boost action of Section 8—already realised in the gravitomagnetic sector that frame-dragging measurements probe. The missing ingredient is the same one Dirac's condition waited on: a physical pair with one member known. Until then the correct classification is a quantisation of permitted pairs, on a quarter lattice, with electromagnetism on its even rungs.

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